

Git & GitHub Revision Notes

1. Initialize a Repository

Create a new Git repository in the current folder:

```
git init
```

Check repository status:

```
git status
```

2. Configure Git Identity

Set username:

```
git config --global user.name "Golu Mishra"
```

Set email:

```
git config --global user.email "golu@thoughtwin.com"
```

Verify configuration:

```
git config --list
```

3. Add Files to Staging Area

Add a specific file:

```
git add first.py
```

Add all files:

```
git add .
```

Check status:

```
git status
```

4. Create Commits

Create a commit:

```
git commit -m "Add first.py"
```

Create a meaningful commit:

```
git commit -m "Create login API"
```

5. Modify Last Commit Message

Correct the last commit message:

```
git commit --amend -m "TeamB: Contact/Index/Login"
```

6. View History

Detailed history:

```
git log
```

Compact history:

```
git log --oneline
```

Graph view:

```
git log --oneline --graph --all
```

Show a specific commit:

```
git show <commit-id>
```

Example:

```
git show a1b2c3d
```

7. Compare Changes

Working directory changes:

```
git diff
```

Staged changes:

```
git diff --staged
```

8. Connect Local Repository to GitHub

Add remote:

```
git remote add origin https://github.com/golu-boop/Python-Training-Week-One.git
```

Check remote:

```
git remote -v
```

Remove remote:

```
git remote remove origin
```

Change remote URL:

```
git remote set-url origin https://github.com/username/repo.git
```

9. Rename Main Branch

Rename current branch to main:

```
git branch -M main
```

10. Push to GitHub

First push:

```
git push -u origin main
```

Normal push:

```
git push
```

Push specific branch:

```
git push origin login
```

Force push (dangerous):

```
git push --force
```

11. Pull Changes

Pull latest changes:

```
git pull
```

Pull from a specific branch:

```
git pull origin main
```

Pull without rebase:

```
git pull origin main --no-rebase
```

Pull unrelated histories:

```
git pull origin main --allow-unrelated-histories
```

12. Branch Management

View branches:

```
git branch
```

Create branch:

```
git branch TeamB
```

Create and switch:

```
git checkout -b login
```

Modern syntax:

```
git switch -c login
```

Switch branch:

```
git switch main
```

or

```
git checkout main
```

13. Merge Branches

Merge login branch:

```
git merge login
```

Merge signout branch:

```
git merge signout
```

Check merged branches:

```
git branch --merged
```

14. Merge Conflicts

When Git cannot automatically merge changes:

Conflict markers:

```
<<<<<< HEAD
Current branch code
=====
Incoming branch code
>>>>>> login
```

Resolve manually by editing the file.

Then:

```
git add .  
git commit -m "Conflict resolved"
```

Push changes:

```
git push
```

Abort merge:

```
git merge --abort
```

15. Delete Branches

Delete local branch:

```
git branch -d TeamB
```

Force delete:

```
git branch -D TeamB
```

Delete remote branch:

```
git push origin --delete TeamB
```

16. Working with Feature Branches

Create login feature:

```
git checkout -b login
```

Commit changes:

```
git add .  
git commit -m "Create login API"
```

Push branch:

```
git push -u origin login
```

Merge into main:

```
git checkout main  
git merge login  
git push origin main
```

17. Useful Troubleshooting Commands

Check repository status:

```
git status
```

Check remote URLs:

```
git remote -v
```

Check commit history:

```
git log --oneline
```

Check HEAD movement history:

```
git reflog
```

List Git commands used:

```
history | grep git
```

Save Git commands to a file:


```
history | grep git > commands.txt
```

18. Daily Workflow

```
git status  
  
git add .  
  
git commit -m "Meaningful commit message"  
  
git push
```

19. Branch Workflow

```
git checkout -b feature-name  
  
# Make changes  
  
git add .  
  
git commit -m "Add feature"  
  
git push -u origin feature-name  
  
git checkout main  
  
git merge feature-name  
  
git push origin main
```

20. Most Important Commands to Memorize

```
git init  
git status  
git add .  
git commit -m ""  
git log --oneline  
git diff
```

```
git remote -v  
git push  
git pull  
git branch  
git switch  
git checkout  
git merge  
git branch -d  
git push origin --delete  
git reflog  
git merge --abort
```